

Git Group Project - One-Page Workflow

1. Clone once

```
git clone <repo-url>
cd <repo-name>
git status
```

2. Start new work

```
git switch main
git pull origin main
git switch -c feature/short-name
```

3. Inspect changes

```
git status
git diff # unstaged
git diff --staged # staged
git diff path/to/file.py
```

4. Stage and commit

```
git add path/to/file.py # selective
git add . # all changes
git diff --staged
git commit -m "Add EWMA volatility targeting"
```

Rule

Make small commits. Each commit should represent one logical change.

5. Push and open PR

```
git push -u origin feature/short-name
# Later: git push
```

PR target

feature/short-name -> main

State what changed, why, how tested, and limitations.

6. Update your branch

```
git fetch origin
git merge origin/main
```

7. Resolve conflicts

```
<<<<<<< HEAD
your version
=====
main version
>>>>>> origin/main
```

Edit to the correct final code. remove markers. then:

```
git add conflicted_file.py
git commit
git push
```

Abort a bad merge

```
git merge --abort
```

8. After PR is merged

```
git switch main
git pull origin main
git branch -d feature/short-name
```

Useful recovery commands

```
git restore file.py # discard unstaged
git restore --staged file.py # unstage only
git stash # save unfinished work
git stash pop # restore it
git log --oneline --graph --decorate
git show <commit-hash>
git revert <commit-hash> # safely undo pushed commit
```

Avoid

git reset --hard and force-pushing unless you know exactly what will be deleted.

Daily checklist

Before editing	git switch main -> git pull -> git switch -c feature/...
Before commit	git status -> git diff -> git add ... -> git diff --staged
Before push	Run code/tests; inspect every staged change
Before merge	PR reviewed; tests pass; conflicts resolved

Always personally inspect **git status**, **git diff**, and **git diff --staged** before letting an AI commit or push.